Youseph Elkhouly

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Education

University of Guelph **September 2023 – April 2027**

Bachelor of Computing, Computer Science Honours
Relevant Coursework: Object-Oriented Programming, Data Structures & Algorithms, Discrete Math, Database Systems, Parallel Programming, Operating Systems, Systems Analysis and Design, Software Reliability & Testing, Software Engineering

Experience

Backboard.io · AI Engineer Intern **May 2026 – Present**

Python · Java · Rust · AWS · Flask · DynamoDB · Backboard SDK · Authlib *Ottawa, Ontario, Canada*

- Shipped **Backboard Studio**, an agentic IDE forked from VS Code, on a 3-person team designing three modes: code editor, a chat-first Simple Mode for non-technical users, and Advanced a kanban orchestrating concurrent Backboard, Claude, and Codex agent sessions.
- Drove the full SDLC: Architected internal systems that offload **80%** of session context/ compresses models memory **3x** and kept interaction latency under **2ms**, validated designs with senior engineers, shipped **150+** commits within two months, and deployed via AWS to **300+** internal/beta users.
- Shipped production features for **Nash**, Backboard's sub product an enterprise chat workspace: Google SSO authentication, Stripe-integrated billing with tiered pricing plans and plan-based token/credit allocation, and API user key minting tied to Backboard's existing platform.

PrepBusiness · Software Engineer Intern **Sep 2025 – Dec 2025**

Python (FastAPI) · TypeScript (React) · SQLite/pgvector · Azure OpenAI API *Hamilton, Ontario*

- Engineered an AI-powered customer support chatbot using FastAPI and retrieval-augmented generation (RAG) with vector embeddings for semantic search across company documentation, implementing async processing and intelligent prompt engineering to automate **65%** of common inquiries
- Developed a chat widget with real-time WebSocket messaging and conversation persistence to enhance user satisfaction through intuitive customer interactions.
- Implemented comprehensive data protection and escalation system with automated PII detection and redaction, coupled with ML confidence scoring to ensure seamless handoff of sensitive or complex cases to human agents, achieving **100%** secure handling of customer data.
- Built a modular AI inference layer with Azure OpenAI API, allowing flexible provider integration

Stylus Marketing · Software Engineer Intern **May 2024 – August 2024**

Python · TypeScript · PostgreSQL · AWS (SQS, Lambda, & EventBridge) · React *Toronto, Ontario, Canada*

- Engineered a serverless inventory tracking and management service adopted by **50+** clients, delivering scalable, cost-efficient operations.
- Architected an automated, event-driven Stripe integration for overdue charge recovery, incorporating dynamic retry logic and real-time alerts to boost payment reliability and accelerate resolution times by **75%**.
- Implemented intuitive dashboards for payment and inventory management, leveraging WebSockets for instant data synchronization and delivering a seamless, user-friendly interface that reduced user onboarding time.

Projects

Frontline - DeltaHacks12 (Hackathon Winner) | **C++ · React · Gemini API · ElevenLabs · Presage · Node.js** ·

- Developed a platform that transforms camera feeds into real-time biometric data and vital signs
- Engineered an ElevenLabs and Gemini-powered first aid voice operator
- Built an ER-ready incident response dashboard that generates standardized SBAR reports, enabling ER workers to review incident video playback and receive patient reports in advance, aiming to reduce diagnosis and handoff

Notipply (Job Scraper/SMS Notification platform) | ***React · Node.js · Express.js · Go · Supabase · Clerk · Stripe · Twilio***

- Built a concurrent Go pipeline to scrape & filter job postings from 1,500+ companies across the U.S. and Canada.
- Built an SMS notification system using the Twilio API to send personalized job alerts based on users' preferences.
- Developed backend services with Supabase and PostgreSQL, & integrated Clerk for user authentication.
- Integrated the Stripe API to manage subscription billing and recurring payments as Notipply grew to 50+ users.

Technical Skills

Languages & Database: Python, Rust, SQL, Javascript, Java, Typescript, PostgreSQL, MongoDB, Firebase
Frameworks & Tools: React, Redux, Express, Spring, Flask, AWS, Docker, Jest, Pytest, Git, CI/CD.